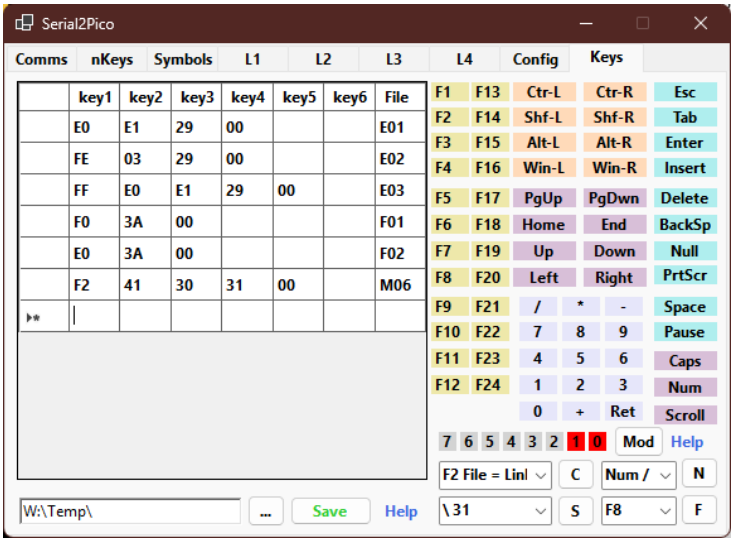
Exercise in using the Keys and Link tabs in the PC App

Construct the following 6 files in the Keys tab of the PC App – press Show Macro Keys in the Config Tab



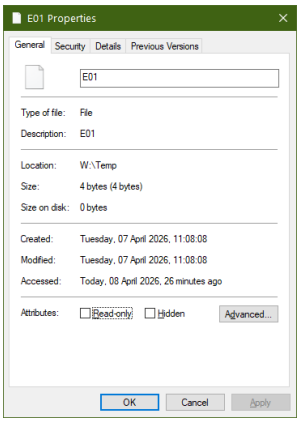
For file E01: Click in first cell of first row (below key1), then click Ctr-L, then click Shf-L, then Esc, then Null. Type E01 into the cell below File(name).
For file E02: Click on the 0 and 1 blocks of the Mod list – they will turn red – that means Ctrl-L and Shf-L are on. Then click on the first cell of the second row and open the list next to the [C] button and select FE 1x Mod byte, then press the button [C] and check if FE shows in the first cell. Then click on the [Mod] button with the two red [1][0] labels next to it and check if 03 is entered in the cell next to FE. Then click on Esc, then Null and name the file E02
For File E03: Click on the first cell of the third row and open the list next to the [C] button and select FF Mod byte, then press the button [C] and check if FF shows in the first cell. Then click Ctr-L, then click Shf-L, then Esc, then Null. Type E03 into the cell below File.
Explanation: File E01 presses the keys Control + Shift + Esc together which opens the Task Manager in Windows. File E01 does the same action but uses the HID modifier bit to indicate that Control + Shift are pressed, namely the value 03 which is 0000 0011, and it then it adds the Esc key value 29 – the Pico uses the flag FE to indicate that the next value 03 is the modifier byte. File E03 also has same action but uses the Pico flag FF to indicate the modifier bit must be constructed from the next two values for the Control and Shift keys, and then it also adds the Esc key value 29. The modifier bit represents the keys Gui-R Alt-R Shf-R Ctr-R Gui-L Alt-L Shf-L Ctr-L in the bits 76543210. The two Pico flags FF and FE are hexadecimal values 0xFe and 0xFF, are not part of the HID protocol, and are only used to signal to the Pico what to do next.

For File F01: Click on the first cell of the fourth row and open the list next to the [C] button and select F0 F1-F24, then press the button [C] and check if F0 shows in the first cell. Open the list next to the [F] button and select F1, then press the button [F] and check if 3A shows in the second cell. Then click on the Null label. Type F01 into the cell below File.
For File F02: Click on the first cell of the fifth row then click on the label Shf-L. Open the list next to the [F] button and select F1, then press the button [F] and check if 3A shows in the second cell. Then click on the Null label. Type F02 into the cell below File.
Explanation: The keys pressed are the function key F1 for file F01, and Shift + F1 for file F02. In Microsoft Word that opens the Help section, and hide and show the ribbon respectively. You only need the flag F0 if the first command is F1-F24. If it is a 2nd or 3rd command you only need the function key F1-F24 value without the F0.

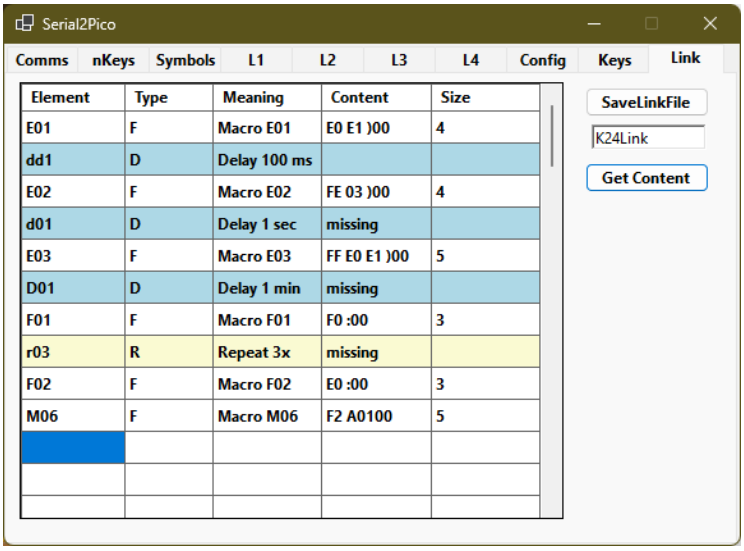
For File M06: Click on the first cell of the sixth row and open the list next to the [C] button and select F2 File=Link, then press the button [C] and check if F2 shows in the first cell. Open the list next to the [F] button and select F8 and press [F], then open the list next to the [S] button and select] 30 and press [S], then and select \ 31 and press [S]. press the label Null. Check that the row has the values F2 41 30 31 00.
Explanation: The flag F2 indicates to the Pico that the contents of the three character filename after the F2 should not be executed but that it contains another filename of which the contents must be executed – i.e. it is a symbolic link file

– similar to the large 100k+ set of nKey files in the Pico macropad. Because that filename must be in ASCII letters – it is A01 – and the keyboard is operating in HID mode in the Keys tab of the PC App, you need to enter the ASCII filename A01 by using the three ASCII letter-codes F8=41=A,]=30=0, and /=31=1 to form the filename A01.

You can now save the six files by pressing the button [Save] – if the path next to the button is empty first select a folder where the files must be saved using the [...] button. Check in the folder if the six files E01, E02, E03, F01, F02, M06 are there – if you have a hex editor, you can check their content as well. You can compare their sizes against what are shown as their content in the keys tab namely 4, 4, 5, 3, 3, and 5 bytes (the null byte at the end adds one byte to the size).



Now open the Link tab in the PC App by clicking the Show Link List in the Config tab – it will at first show an empty list:

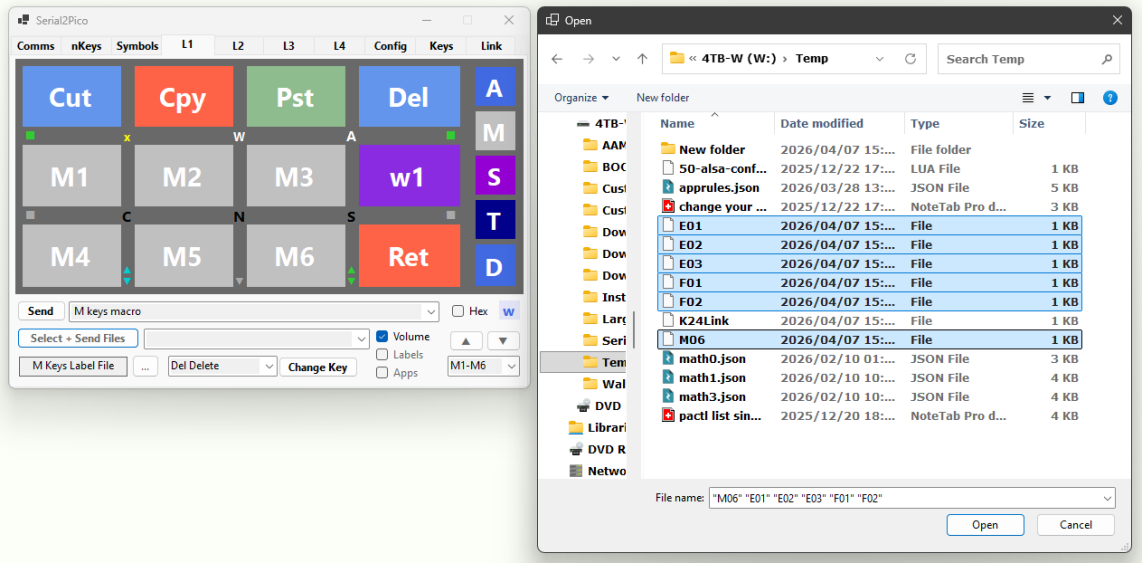


Element	Type	Meaning	Content	Size
E01	F	Macro E01	E0 E1)00	4
dd1	D	Delay 100 ms		
E02	F	Macro E02	FE 03)00	4
d01	D	Delay 1 sec	missing	
E03	F	Macro E03	FF E0 E1)00	5
D01	D	Delay 1 min	missing	
F01	F	Macro F01	F0 :00	3
r03	R	Repeat 3x	missing	
F02	F	Macro F02	E0 :00	3
M06	F	Macro M06	F2 A0100	5

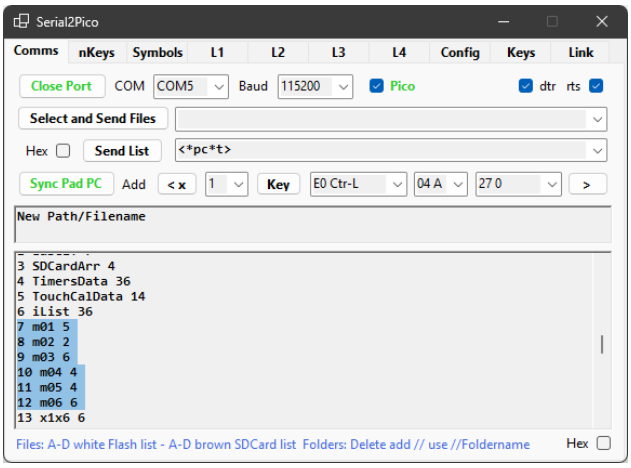
Start by typing in the first filename E01 in the first row below Element, then type in dd1, the third row has the file E02, then type d01, then type E03, D01, F01, r03, F02, and M06. Then press the [Get Content] button and check if the values are the same as what are shown above. Note that the file M06 has the filename A01 in its content. Type in a Link filename such as K24Link in the textbox below the button [SaveLinkFile], and press the button [SaveLinkFile] which will save the list file that you have produced as file K24Link which is a plain text file with the content; E01dd1E02d01E03D01F01r03F02M06.

Explanation: The file K24Link will execute the 3-character commands one after the other. First file E01, then it will wait for I00 milliseconds, then execute file E02 and wait for 1 second, and the same type of actions further on in the list. The repeat r03 means it will do the action in file M02 three times (i.e. it opens, closes and re-opens the Word ribbon).

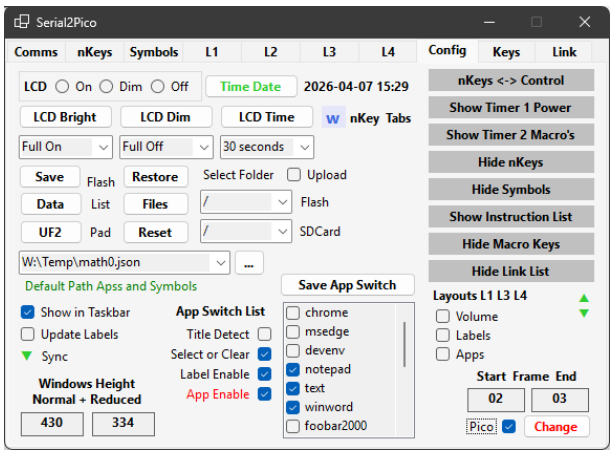
You can now upload the files to the Pico macro-pad:

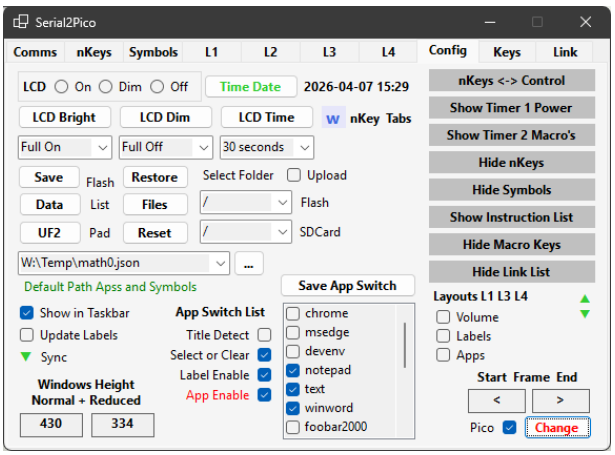
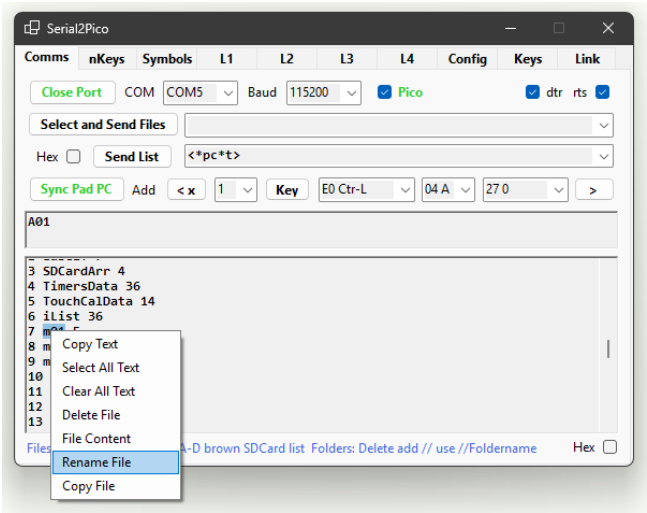
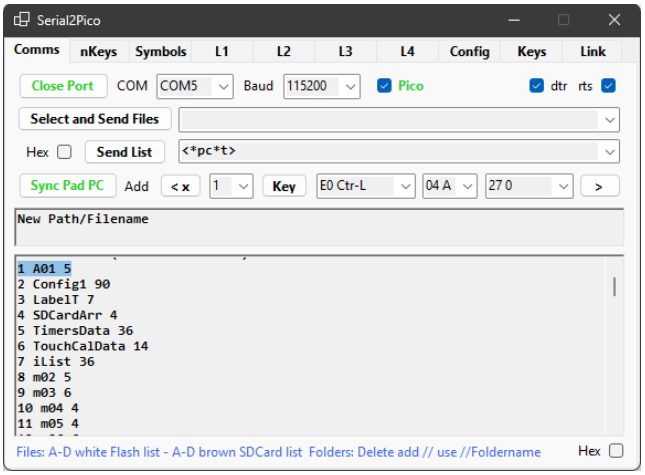
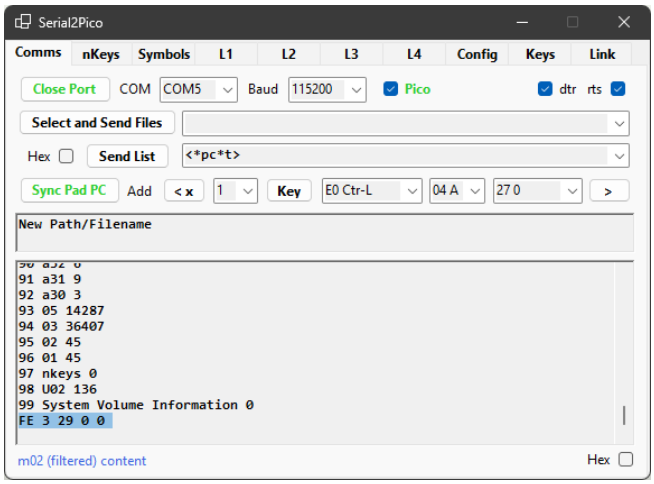
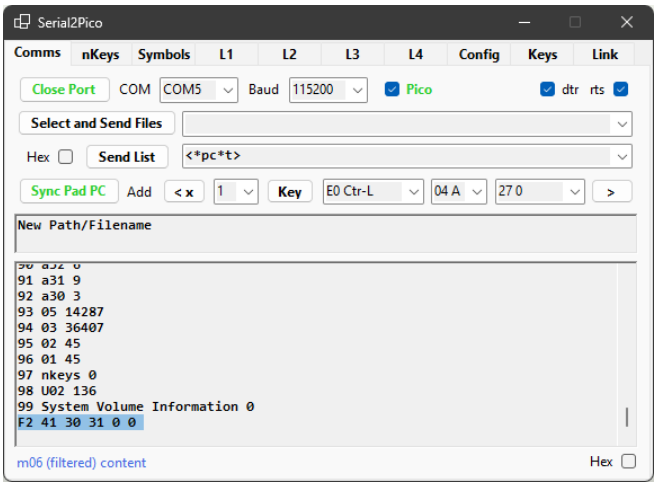


Open L1 – the M keys and press [Select + Send Files], then open the folder where your six files are located and select them then press [Open] in the Windows file explorer, and the files will then be copied as m01 to m06 to either the SDCard if you have a brown A-D or Flash if it is a white A-D. *If you had page B open of the M keys i.e. M7 to M12, then the files copied would be named m07 to m12 – and the same for pages C and D as files m13 - m18, and m19 - m24.*



You will notice that saving the files have added one null byte at the end, but importantly, file m02 which is file E02 on the PC has been *truncated to 2 bytes*. The reason is that the macropad and PC App, are using framing stop and start bytes of 02 and 03, therefore the Pico saw the 03 modifier byte and used it as a stop byte to truncate and save the file. To get around this for file E02 change the stop and start bytes to < > in the Config tab (check the Pico macropad for a confirmation message after the change):





Then press the button [Change], and re-upload all six files. (If you only uploaded the one file E02 it willl get the name m01.) Now make sure you are in the same storage medium as where your six files are saved – if Flash the white A-D, if SD Card then brown A-D. You can now test the files by pressing keys m01, m02, m03, m04 and m05. For m04, and m05 have MSWord open when testing the keys. For key m06 to work it needs a file A01 – rename file m01 to A01 by typing the name A01 in the textbox above the file list, then select the letters m01, right-click m01 and select rename. Then you can press m06 - it should execute the contents of file A01 which is to open the Task Manager. Remember that the Flash filesystem is case-sensitive so files a01 and A01 are two different files – this is not the case for the SD Card.